Chapter 1. Linear Regression and Gradient Descent

Regression, normal equations, gradient descent, generalization

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1 The Regression Problem

We observe $(\mathbf{x}_i, y_i)$, $i = 1, \dots, n$, with $\mathbf{x}_i \in \mathbb{R}^k$, $y_i \in \mathbb{R}$, and seek a function predicting y from $\mathbf{x}$. The "best line" minimizes the sum of squared errors (SSE): rotating a benchmark line traces an SSE curve whose minimizer is the fit.

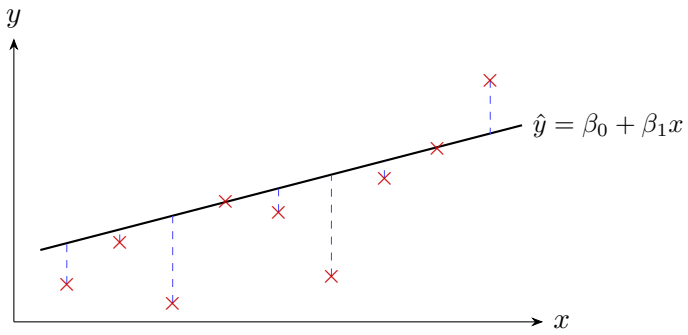
Definition 1.1: linear regression model and design matrix

$$y_i = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij} + \varepsilon_i, \qquad i = 1, \dots, n.$$

Absorb the intercept with a constant input 1: let $d = k + 1$, $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)^\top \in \mathbb{R}^d$, and let the **design matrix** $X \in \mathbb{R}^{n \times d}$ have ith row $(1, x_{i1}, \dots, x_{ik})$. Then

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad f(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2 = (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}).$$

Rows of X index samples; columns index features.



Residuals $y_i - \hat{y}_i$ (dashed); least squares minimizes the sum of their squares.

2 Matrix Calculus Toolkit

Motivation. A few gradient identities for linear and quadratic forms make every loss computation in the course one line.

Convention: the gradient of a scalar with respect to a vector is a column vector of the same size.

Theorem 1.2: basic gradients

For $\mathbf{a} \in \mathbb{R}^d$ and $A \in \mathbb{R}^{d \times d}$,

$$\nabla_{\boldsymbol{\beta}}(\mathbf{a}^\top \boldsymbol{\beta}) = \mathbf{a}, \qquad \nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = (A + A^\top)\boldsymbol{\beta}, \qquad \nabla^2_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = A + A^\top.$$

If A is symmetric the last two become $2A\boldsymbol{\beta}$ and $2A$.

Proof. $\partial(\sum_j a_j \beta_j)/\partial \beta_k = a_k$. For $\boldsymbol{\beta}^\top A \boldsymbol{\beta} = \sum_{i,j} A_{ij} \beta_i \beta_j$ the product rule picks up the terms $i = k$ and $j = k$:

$$\frac{\partial}{\partial \beta_k} \sum_{i,j} A_{ij} \beta_i \beta_j = \sum_j A_{kj} \beta_j + \sum_i A_{ik} \beta_i = (A\boldsymbol{\beta})_k + (A^\top \boldsymbol{\beta})_k.$$

Differentiating again gives $A + A^\top$. ∎

3 Normal Equations

Motivation. Least squares has a closed-form solution: projection of $\mathbf{y}$ onto the column space of X.

Theorem 1.3: least squares solution

$\nabla f(\boldsymbol{\beta}) = 2X^\top(X\boldsymbol{\beta} - \mathbf{y})$ and $\nabla^2 f = 2X^\top X \succeq 0$. Every minimizer solves the **normal equations**

$$X^\top X\boldsymbol{\beta} = X^\top \mathbf{y}.$$

If X has full column rank d, then $X^\top X \succ 0$ and the unique minimizer is $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top \mathbf{y}$.

Proof. $f = \mathbf{y}^\top\mathbf{y} - 2\mathbf{y}^\top X\boldsymbol{\beta} + \boldsymbol{\beta}^\top X^\top X\boldsymbol{\beta}$, so $\nabla f = -2X^\top\mathbf{y} + 2X^\top X\boldsymbol{\beta}$, $\nabla^2 f = 2X^\top X$. Since $\mathbf{v}^\top X^\top X\mathbf{v} = \|X\mathbf{v}\|^2 \geq 0$, f is convex and stationary points are global minimizers. If the columns of X are independent, $X\mathbf{v} \neq \mathbf{0}$ for $\mathbf{v} \neq \mathbf{0}$, so $X^\top X \succ 0$ is invertible, f is strictly convex and the minimizer is unique. ■

Remark 1.4: geometry and rank

$X^\top(\mathbf{y} - X\hat{\boldsymbol{\beta}}) = \mathbf{0}$: $X\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathrm{col}(X)$, and residuals sum to zero (first column $\mathbf{1}_n$). With dependent columns ($n > d$ does not prevent this) the pseudoinverse gives the minimum-norm solution; numerically use QR or SVD.

Example 1.5: least squares on three points

Points $(0,1),(1,2),(2,2)$ and model $y = \beta_0 + \beta_1 x$:

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad X^\top X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad X^\top\mathbf{y} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad (X^\top X)^{-1} = \frac{1}{6}\begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix}.$$

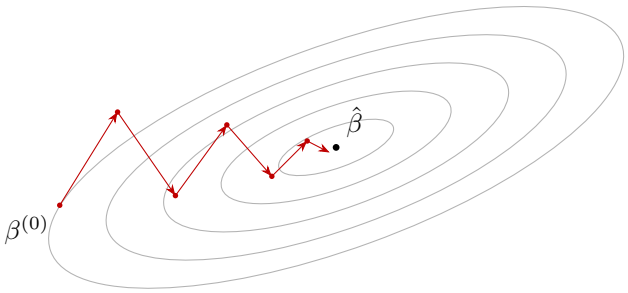
Hence $\hat{\boldsymbol{\beta}} = \frac{1}{6}(25 - 18, -15 + 18)^\top = (7/6,\ 1/2)^\top$; predictions $(7/6, 5/3, 13/6)$, residuals $(-1/6, 1/3, -1/6)$, SSE $= 1/6$.

4 Gradient Descent

4.1 The algorithm

$$\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta\nabla f(\boldsymbol{\beta}_t) = \boldsymbol{\beta}_t + 2\eta X^\top(\mathbf{y} - X\boldsymbol{\beta}_t),$$

i.e. $\beta_j \leftarrow \beta_j + 2\eta\sum_i (y_i - \hat{y}_i)x_{ij}$ with $x_{i0} = 1$ (the 2 is often absorbed into η); each full-batch step uses all samples. By Cauchy–Schwarz, $\nabla f^\top\mathbf{v}$ over unit $\mathbf{v}$ is minimized at $\mathbf{v} = -\nabla f/\|\nabla f\|$ (steepest descent).



Gradient descent on an elongated quadratic bowl zigzags toward $\hat{\beta}$.

4.2 Convergence on least squares

Let $J(\boldsymbol{\beta}) = \frac{1}{2}\|X\boldsymbol{\beta} - \mathbf{y}\|^2$, $\nabla J = X^\top(X\boldsymbol{\beta} - \mathbf{y})$, $H = X^\top X \succ 0$.

Theorem 1.6: exact convergence condition

Gradient descent $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta\nabla J(\boldsymbol{\beta}_t)$ converges to $\hat{\boldsymbol{\beta}}$ for every initial point if and only if

$$0 < \eta < \frac{2}{\lambda_{\max}(X^\top X)}.$$

Proof. Error recursion. $X^\top\mathbf{y} = H\hat{\boldsymbol{\beta}}$, so $\nabla J(\boldsymbol{\beta}_t) = H(\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}})$ and, with $\mathbf{e}_t = \boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}$,

$$\mathbf{e}_{t+1} = \mathbf{e}_t - \eta H\mathbf{e}_t = (\mathbf{I} - \eta H)\mathbf{e}_t, \qquad \mathbf{e}_t = (\mathbf{I} - \eta H)^t\mathbf{e}_0.$$

Diagonalize. $H = Q\Lambda Q^\top$, Q orthogonal, $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$, $\lambda_i > 0$; $\mathbf{I} - \eta H = Q(\mathbf{I} - \eta\Lambda)Q^\top$. With $\mathbf{c}_t = Q^\top\mathbf{e}_t$, $\mathbf{c}_{t+1} = (\mathbf{I} - \eta\Lambda)\mathbf{c}_t$:

$$c_{t,i} = (1 - \eta\lambda_i)^t\, c_{0,i}.$$

As $\|\mathbf{c}_t\| = \|\mathbf{e}_t\|$, $\mathbf{e}_t \to \mathbf{0}$ iff every $c_{t,i} \to 0$.

Sufficiency. If $|1 - \eta\lambda_i| < 1$ for all i, every coordinate tends to zero for every $\mathbf{e}_0$.

Necessity. If $|1 - \eta\lambda_i| \geq 1$ for some i, take $\mathbf{e}_0 = \mathbf{q}_i$: $\|\mathbf{e}_t\| = |1 - \eta\lambda_i|^t \geq 1$.

Solve. $|1 - \eta\lambda_i| < 1 \iff 0 < \eta < 2/\lambda_i$; for all i this is $0 < \eta < 2/\lambda_{\max}$. ■

Remark 1.7: conditioning

$\max_i |1 - \eta\lambda_i|$ is minimized by $\eta^\star = 2/(\lambda_{\max} + \lambda_{\min})$, rate $(\kappa - 1)/(\kappa + 1)$, $\kappa = \lambda_{\max}/\lambda_{\min}$: ill-conditioning slows gradient descent (Chapter 6). At $\eta = 2/\lambda_{\max}$ the top direction oscillates with factor -1, so the inequality is strict. In Example 1.5, $\lambda(X^\top X) = 4 \pm \sqrt{10}$, so $0 < \eta < (4 - \sqrt{10})/3 \approx 0.279$.

5 Polynomial Features and Generalization

5.1 Feature maps

$\boldsymbol{\phi}_M(x) = (1, x, x^2, \ldots, x^M)^\top$, $y(x) = \sum_{j=0}^M \beta_j x^j$: nonlinear in x, linear in $\boldsymbol{\beta}$, with Vandermonde-type design matrix of rows $\boldsymbol{\phi}_M(x_i)^\top$.

5.2 Training error versus generalization

Nested models make the minimal training SSE nonincreasing in M, but validation error can rise: high degree fits noise (overfitting), low degree misses the trend (underfitting). Metric:

$$\mathrm{MSE} = \frac{1}{n}\mathrm{SSE}, \qquad E_{\mathrm{RMS}} = \sqrt{\mathrm{MSE}}.$$

Remark 1.8: roles of the data splits

Training data fit coefficients, validation data choose degree and regularization, a test set estimates final performance. Degree 9 overfits badly with $n = 10$ but not with $n = 100$; "ten points per parameter" is only a heuristic. Regularization (Chapter 3) is the second remedy.

6 Proof Supplement: Uniqueness and the Pseudoinverse

Motivation. When $X^\top X$ is singular, the pseudoinverse picks the minimum-norm solution, the one gradient descent from zero finds.

Theorem 1.9: set of minimizers

The minimizers of $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$ are exactly the solutions of $X^\top X\boldsymbol{\beta} = X^\top \mathbf{y}$; this system is always consistent, and the solution set is $\hat{\boldsymbol{\beta}}_0 + \ker X$ for any particular solution $\hat{\boldsymbol{\beta}}_0$. Among them, $X^+\mathbf{y}$ has minimal Euclidean norm.

Proof. Consistency: $X^\top X\mathbf{v} = \mathbf{0} \Rightarrow \|X\mathbf{v}\|^2 = 0$, so $\mathrm{col}(X^\top)$ and $\mathrm{col}(X^\top X)$ both have complement $\ker X$ and coincide; $X^\top \mathbf{y}$ lies in it. Two solutions differ by $\mathbf{v} \in \ker X$ and give the same fit. With thin SVD $X = U\Sigma V^\top$ (r positive singular values), every solution is $V_r\Sigma_r^{-1}U_r^\top\mathbf{y} + \mathbf{v}$, $\mathbf{v} \perp \mathrm{range}(V_r)$; by Pythagoras the norm is minimal at $\mathbf{v} = \mathbf{0}$, i.e. $X^+\mathbf{y}$. ■

7 Selected exercises

Motivation. Three derivations: the hat matrix, gradient descent's minimum-norm limit, and the optimal step size.

Exercise 1.1 (Hat matrix and the residual sum of squares)

Let $X \in \mathbb{R}^{n\times d}$ have full column rank $d < n$ and put $H = X(X^\top X)^{-1}X^\top$.
(a) Show that H is symmetric and idempotent, $HX = X$ and $\mathrm{tr}\, H = d$; show that $\mathbf{I} - H$ is symmetric and idempotent with $\mathrm{tr}(\mathbf{I} - H) = n - d$.
(b) Show that the fitted values are $H\mathbf{y}$, the residual vector is $\mathbf{e} = (\mathbf{I}-H)\mathbf{y}$, $\mathrm{SSE} = \mathbf{y}^\top(\mathbf{I}-H)\mathbf{y}$, and $\|\mathbf{y}\|^2 = \|H\mathbf{y}\|^2 + \|\mathbf{e}\|^2$.
(c) Let $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$. Show that $\mathbb{E}[\mathrm{SSE}] = (n - d)\sigma^2$, so $\mathrm{SSE}/(n - d)$ is unbiased for σ^2.

Solution. [Supplement]

(a) $(X^\top X)^{-1}$ is symmetric, so $H^\top = H$, and

$$H^2 = X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1}X^\top = H, \qquad HX = X(X^\top X)^{-1}X^\top X = X.$$

$\mathrm{tr}\, H = \mathrm{tr}\,\big((X^\top X)^{-1}X^\top X\big) = \mathrm{tr}\,\mathbf{I}_d = d$. Then $\mathbf{I} - H$ is symmetric, $(\mathbf{I} - H)^2 = \mathbf{I} - 2H + H^2 = \mathbf{I} - H$, $\mathrm{tr}(\mathbf{I} - H) = n - d$.

(b) $X\hat{\boldsymbol{\beta}} = X(X^\top X)^{-1}X^\top \mathbf{y} = H\mathbf{y}$, so $\mathbf{e} = (\mathbf{I} - H)\mathbf{y}$ and by (a)

$$\mathrm{SSE} = \mathbf{e}^\top \mathbf{e} = \mathbf{y}^\top (\mathbf{I} - H)^\top (\mathbf{I} - H)\mathbf{y} = \mathbf{y}^\top (\mathbf{I} - H)\mathbf{y}.$$

$(H\mathbf{y})^\top \mathbf{e} = \mathbf{y}^\top (H - H^2)\mathbf{y} = 0$, so expanding $\|H\mathbf{y} + \mathbf{e}\|^2$ gives the identity.

(c) $(\mathbf{I} - H)X = 0$ gives $\mathbf{e} = (\mathbf{I} - H)\boldsymbol{\varepsilon}$ and $\mathrm{SSE} = \boldsymbol{\varepsilon}^\top (\mathbf{I} - H)\boldsymbol{\varepsilon}$. For fixed A,

$$\mathbb{E}[\boldsymbol{\varepsilon}^\top A \boldsymbol{\varepsilon}] = \mathbb{E}\,\mathrm{tr}(A\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top) = \mathrm{tr}\,(A\,\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top]) = \sigma^2\,\mathrm{tr}\,A.$$

With $A = \mathbf{I} - H$ this is $(n - d)\sigma^2$; hence the MLE SSE/n (Chapter 3) is biased downward. ■

Exercise 1.2 (Gradient descent from zero finds $X^+\mathbf{y}$)

Let $X \neq 0$ have rank r and thin SVD $X = U_r \Sigma_r V_r^\top$ with singular values $s_1, \ldots, s_r > 0$; let the columns of V_0 complete those of V_r to an orthonormal basis of $\mathbb{R}^d$. Run gradient descent on $J(\boldsymbol{\beta}) = \frac{1}{2}\|X\boldsymbol{\beta} - \mathbf{y}\|^2$ with $0 < \eta < 2/\lambda_{\max}(X^\top X)$; $X^\top X$ need not be invertible.
(a) Show that if $\boldsymbol{\beta}_0 = \mathbf{0}$, every iterate lies in $\mathrm{range}(X^\top) = \mathrm{range}(V_r)$.
(b) Show that if $\boldsymbol{\beta}_0 = \mathbf{0}$, then $\boldsymbol{\beta}_t \to X^+\mathbf{y} = V_r \Sigma_r^{-1} U_r^\top \mathbf{y}$.
(c) Show that for a general $\boldsymbol{\beta}_0$ the limit is $X^+\mathbf{y} + V_0 V_0^\top \boldsymbol{\beta}_0$.

Solution. [Supplement]

(a) $\nabla J(\boldsymbol{\beta}) = X^\top(X\boldsymbol{\beta} - \mathbf{y}) \in \mathrm{range}(X^\top)$, so by induction from $\boldsymbol{\beta}_0 = \mathbf{0}$ every $\boldsymbol{\beta}_t$ stays in this subspace. $X^\top = V_r \Sigma_r U_r^\top$ with $\Sigma_r U_r^\top$ of full row rank, so $\mathrm{range}(X^\top) = \mathrm{range}(V_r)$.

(b) *Step 1 (coordinates).* By (a), $\boldsymbol{\beta}_t = V_r \mathbf{c}_t$, $\mathbf{c}_t = V_r^\top \boldsymbol{\beta}_t$. With $X^\top X = V_r \Sigma_r^2 V_r^\top$, $X^\top \mathbf{y} = V_r \Sigma_r U_r^\top \mathbf{y}$,

$$\mathbf{c}_{t+1} = \mathbf{c}_t - \eta(\Sigma_r^2 \mathbf{c}_t - \Sigma_r U_r^\top \mathbf{y}).$$

Step 2 (error recursion). With $\mathbf{c}^\star = \Sigma_r^{-1} U_r^\top \mathbf{y}$, $\mathbf{c}_{t+1} - \mathbf{c}^\star = (\mathbf{I} - \eta\Sigma_r^2)(\mathbf{c}_t - \mathbf{c}^\star)$, diagonal with factors $1 - \eta s_i^2$.

Step 3 (contraction). s_i^2 are the nonzero eigenvalues of $X^\top X$, so $0 < \eta s_i^2 < 2$, $|1 - \eta s_i^2| < 1$, $\mathbf{c}_t \to \mathbf{c}^\star$ and $\boldsymbol{\beta}_t \to V_r \Sigma_r^{-1} U_r^\top \mathbf{y} = X^+\mathbf{y}$.

(c) Write $\boldsymbol{\beta}_t = V_r \mathbf{c}_t + V_0 \mathbf{d}_t$. As $\nabla J \in \mathrm{range}(V_r) \perp V_0$, $\mathbf{d}_t = V_0^\top \boldsymbol{\beta}_0$ for all t. Since $X^\top X V_0 = 0$, $\mathbf{c}_t$ obeys Step 1 from a different start, and Steps 2–3 give $\mathbf{c}_t \to \mathbf{c}^\star$, so $\boldsymbol{\beta}_t \to X^+\mathbf{y} + V_0 V_0^\top \boldsymbol{\beta}_0$: the $\ker X$ component never moves (implicit regularization). ■

Exercise 1.3 (The optimal constant step size)

Let $H = X^\top X \succ 0$ have eigenvalues $0 < \lambda_{\min} = \lambda_d \leq \cdots \leq \lambda_1 = \lambda_{\max}$, $\kappa = \lambda_{\max}/\lambda_{\min}$, and $\rho(\eta) = \max_i |1 - \eta\lambda_i|$ for $\eta > 0$.
(a) Show that $\rho(\eta) = \max\{|1 - \eta\lambda_{\min}|, |1 - \eta\lambda_{\max}|\}$.
(b) Show that ρ is minimized at $\eta^\star = 2/(\lambda_{\max} + \lambda_{\min})$ and $\rho(\eta^\star) = (\kappa - 1)/(\kappa + 1)$.
(c) Deduce $\|\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}\| \leq \rho(\eta)^t \|\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}}\|$, and compute $\eta^\star$ and $\rho(\eta^\star)$ for Example 1.5.

Solution. [Supplement]

(a) $\lambda \mapsto |1 - \eta\lambda|$ is convex, so its maximum on $[\lambda_{\min}, \lambda_{\max}]$, which contains all λ_i and has eigenvalue endpoints, is at an endpoint.

(b) *Step 1 (lower bound).* By (a), $\rho(\eta) \geq m(\eta) := \max\{1 - \eta\lambda_{\min},\ \eta\lambda_{\max} - 1\}$. The first is decreasing, the second increasing, and they meet at $\eta^\star$, so $m(\eta) \geq m(\eta^\star)$ with

$$m(\eta^\star) = 1 - \frac{2\lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\kappa - 1}{\kappa + 1}.$$

Step 2 (attained). $1-\eta^\star\lambda_{\min}=\frac{\kappa-1}{\kappa+1}\geq 0$ and $1-\eta^\star\lambda_{\max}=-\frac{\kappa-1}{\kappa+1}$, so $\rho(\eta^\star)=m(\eta^\star)\leq\rho(\eta)$ for all $\eta>0$.

(c) From the convergence proof, $|c_{t,i}|=|1-\eta\lambda_i|^t|c_{0,i}|\leq\rho(\eta)^t|c_{0,i}|$; square, sum, and use $\|\mathbf{c}_t\|=\|\mathbf{e}_t\|$. In Example 1.5, $\lambda_{\max,\min}=4\pm\sqrt{10}$, so $\eta^\star=1/4$ and

$$\rho(\eta^\star)=\frac{\lambda_{\max}-\lambda_{\min}}{\lambda_{\max}+\lambda_{\min}}=\frac{2\sqrt{10}}{8}=\frac{\sqrt{10}}{4}\approx 0.791,$$

only about 21% per step, since $\kappa\approx 8.6$. ■